

```
public static int task6CM(int[] input) {  
    int n = input.length;  
    int maxLength = 0;  
    int[] dp = new int[n];  
    for (int i = 0; i < n; i++) {  
        dp[i] = 1;  
        for (int j = 0; j < i; j++) {  
            if (input[i] > input[j]) {  
                dp[i] = Math.max(dp[i], dp[j] + 1);  
            }  
        }  
    }  
    for (int length : dp) {  
        maxLength = Math.max(maxLength, length);  
    }  
    return maxLength;  
}  
  
public static void main(String[] args) {  
    int[] input = {10, 9, 2, 5, 3, 7, 101, 18};  
    int result = task6CM(input);  
    System.out.println(result);  
}
```